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Department of Electronics and Telecommunication Engineering

Summer Semester Examination May-2015

Subject: Digital Communication

Date: 13/05/2015

Time: 2.00pm - 5.00 pm

Semester: VI

Max. Marks: 70

Instructions to the students:-

1. Question No.1 is compulsory. Attempt any FIVE questions from Question No. 2 to Question No. 7. Total number of questions to attempt =6.
2. Assume suitable data wherever necessary and mention it clearly.

Q.1 Solve.

[10]

- I. The Nyquist rate for a signal $f(t) = 10 \cos(50\pi t) \cos^2(150\pi t)$ in samples per second is _____.
a) 150 b) 200 c) 300 d) 600
- II. Quantization noise can be reduced by increasing _____.
a) Bandwidth b) Sampling rate
c) No. of quantization levels d) all of these
- III. Two dice are thrown simultaneously. The probability that the sum is 7 is _____.
a) 1/7 b) 1/6 c) 7/36 d) 1/36
- IV. The autocorrelation function of white noise is _____.
a) A constant b) Gaussian
c) Delta function d) $\exp(-|\tau|)$
- V. The processing gain of DS/BPSK spread spectrum system is 1024. The Number of shift register stages needed to generate the PN sequence is _____.
a) 8 b) 10 c) 12 d) 5
- VI. Information content in a universally true event is _____.
a) Infinite b) zero c) positive const. d) negative const.
- VII. Area under the pdf curve is _____.
a) 0 b) 1 c) ∞ d) none of these
- VIII. Effect of noise is more in _____.
a) ASK b) FSK c) BPSK d) QPSK
- IX. For a Noise-free channel Mutual information is equal to _____.
a) $H(x)$ b) $H(y)$ c) $H(x,y)$ d) All of these
- X. Optimum filter is known as matched filter when input noise is _____.
a) Shot noise b) white noise c) thermal noise d) all of these

- Q.2 A)** A seven segment display emits a 0 with a probability 0.25, a 1 with a probability of 0.2, 2 with a probability of 0.15, 3,4,5 and 6 with a probability of 0.09, 7 and 8 with a probability of 0.015 each and 9 with a probability of 0.01. [6]
- i. Find the efficiency of the Shannon-Fano code for this.
 - ii. Find the avg. information of the source. Compare your results with average information calculated when the display emits all the digits with equal probability.

- B) Explain with neat diagram. [6]
 i. Aliasing effect
 ii. Aperture effect

- Q.3 A) Find mean and variance of the random variable x whose pdf is given by [4]

$$f_x(x) = \frac{3}{32} (-x^2 + 8x - 12) \quad \text{for } 2 \leq x < 6$$

$$0 \quad \text{elsewhere}$$

- B) Define with suitable mathematical expressions [4]

- i) Random Variable ii) Mean
 iii) Variance iv) Standard deviation

- C) The length of time x , needed by students in a particular course to complete a 1 hour examination is a random variable with pdf given by

$$f(x) = k(x^2 + x) \quad \text{if } 0 \leq x \leq 1,$$

$$= 0 \quad \text{else}$$

For the random variable x , [4]

- i. Find the value that makes $f(x)$, a probability density function (pdf).
 ii. Find the cumulative distribution function (CDF).
 iii. Graph for CDF and pdf.
 iv. Find the probability that a randomly selected student will finish the exam in less than half an hour. <https://www.batuonline.com>

- Q.4 A) Derive the expression for signal to quantization noise ratio for PCM. Modify the same for sinusoidal signal. [6]
 B) Explain the quantization in brief. What are disadvantages of uniform quantization? [3]
 C) Write a short note on Delta Modulation. [3]

- Q.5 A) What is noise? State and explain types of noise in communication channel. [4]
 B) Write a note on intersymbol interference. [2]
 C) What is meant by matched filter? Derive the equation for impulse response of it. [6]

- Q.6 A) Derive the expression for probability of error for a BPSK system. [3]
 B) Describe with the block diagram and waveforms, QPSK modulation. [5]
 C) Write a note on constellation diagram. Draw the same for 8-PSK, 16-QAM. [4]

- Q.7 A) The $(1/2, 3)$ convolutional encoder has generating vectors as $g_1 = (100)$ and $g_2 = (110)$. Sketch the encoder and trellis diagram. Decode the sequence 11100000101 using Viterbi algorithm. [6]
 B) Explain spread spectrum modulation. Define chip and processing gain parameters. State advantages and application of the same. [6]

-----ALL THE BEST-----